

On Efficient Quantum Block Encoding

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Abstract

1 Preliminaries and notations

1.1 Notations

We adopt the commonly used notation for binary numbers: for an integer power of two $P = 2^p$ and any $y \in [P]$ if $y = (y_{p-1}y_{p-2} \cdots y_0)$ in the binary system, the corresponding quantum state is $|y\rangle \equiv |y_{p-1} \cdots y_1 y_0\rangle$.

This extends to an m -tuple $\mathbf{x} = (x_1, \cdots, x_m)$ with $x_j \in [P]$. The corresponding quantum state is given by $|x_m\rangle \cdots |x_1\rangle$, where $|x_j\rangle = |x_{j,p-1} \cdots x_{j,0}\rangle, \forall j$. For a multivariate function $g : [P]^m \rightarrow \mathbb{R}$, we denote by D_g the diagonal multiplication operator on $\mathbb{C}^{mp}$:

$$|x_m\rangle \cdots |x_1\rangle \mapsto g(x_1, \cdots, x_m) |x_m\rangle \cdots |x_1\rangle. \quad (1)$$

For a vector $\mathbf{v} = (v_1, v_2, \cdots, v_m)$, we denote $\text{diag}(\mathbf{v})$ as the diagonal matrix $\text{diag}(v_1, v_2, \cdots, v_m)$.

1.2 Discretization of pseudo-differential operators

As mentioned in section intro, the PDO considered in this paper is defined for periodic functions on $\Omega = [0, 1]^d$:

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in \mathbb{Z}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} a(\mathbf{x}, \boldsymbol{\xi}) \hat{f}(\boldsymbol{\xi}).$$

In most numerical treatments, the function f is given on a discrete grid

$$X = \left\{ \frac{\mathbf{x}}{P} \equiv \left(\frac{x_1}{P}, \cdots, \frac{x_d}{P} \right) : x_j \in [P] \right\},$$

where $P = 2^p$ is the number of discrete points used for each dimension. Since the space variable takes value on the Cartesian grid X , the frequency domain is discretized correspondingly on $\{-P/2, \cdots, P/2 - 1\}^d$, which leads to the discretized PDO:

$$Af(\mathbf{x}) \equiv \sum_{\boldsymbol{\xi} \in \{-P/2, \cdots, P/2 - 1\}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} a\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (2)$$

Though the frequency variable $\boldsymbol{\xi}$ is discretized on $\{-P/2, \cdots, P/2 - 1\}^d$ in (2), by the convention of DFT, the frequency $(P/2, \cdots, P - 1)$ is often identified with $(-P/2, \cdots, -1)$, respectively, since P is a period for the frequency variable after DFT. In other words, the discretized PDO can be written as

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \tilde{a}\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d,$$

where

$$\tilde{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv a \left(\mathbf{x}, \boldsymbol{\xi} - P \sum_{\xi_j \geq P/2} \mathbf{e}_j \right),$$

and $\mathbf{e}_j$ is the j -th standard basis vector in $\mathbb{C}^d$. To simplify the notation and avoid repetitive use of P , we further define

$$\check{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv \tilde{a} \left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi} \right),$$

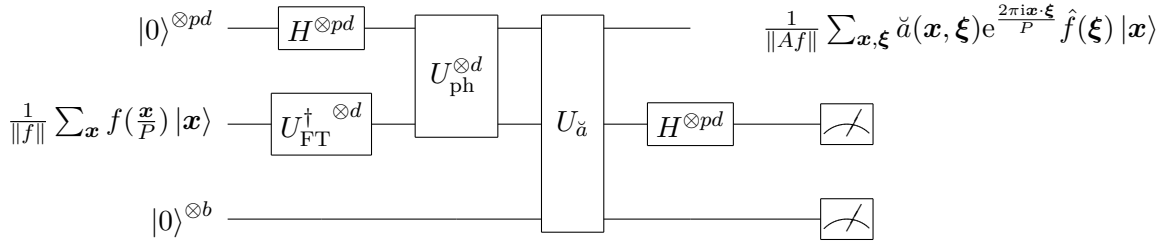
and the discretized PDO becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \check{a}(\mathbf{x}, \boldsymbol{\xi}) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (3)$$

It is clear that $\sup |\check{a}| = \sup |a|$. We also refer to $\check{a}$ as the symbol of the PDO to be computed.

2 Block encoding for generic symbols

This section is concerned with the block encoding of the PDO (3) with a generic symbol $a(\mathbf{x}, \boldsymbol{\xi})$, without assuming any additional structure. In order to compute the PDO in (3), a simple strategy is to first lift the state to the phase space $[P]^d \times [P]^d$. Then the multiplication of $a(\mathbf{x}, \boldsymbol{\xi})$ in (3) can be performed by diagonal matrix block encodings. We can construct the following entire circuit.



Now we explain the above circuit in more details. First, we assume that the information of the function f is prepared by a normalized vector $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}\rangle$, where $\|f\|$ is the normalization factor

$$\|f\| = \sqrt{\sum_{\mathbf{x} \in [P]^d} \left| f\left(\frac{\mathbf{x}}{P}\right) \right|^2}.$$

For the rest of the paper, we assume the accessibility of the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}\rangle$ as an input.

Step 1. Apply QFT and lift the input state to the phase space. We first obtain the representation of f in the frequency domain by QFT. After applying the IQFT to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}\rangle$ for d times, we get

$$\frac{1}{\|f\|} \frac{1}{\sqrt{P^d}} \sum_{\boldsymbol{\xi}} \sum_{\mathbf{x}} f\left(\frac{\mathbf{x}}{P}\right) e^{-2\pi i \boldsymbol{\xi} \cdot \mathbf{x} / P} |\boldsymbol{\xi}\rangle = \frac{\sqrt{P^d}}{\|f\|} \sum_{\boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\boldsymbol{\xi}\rangle. \quad (4)$$

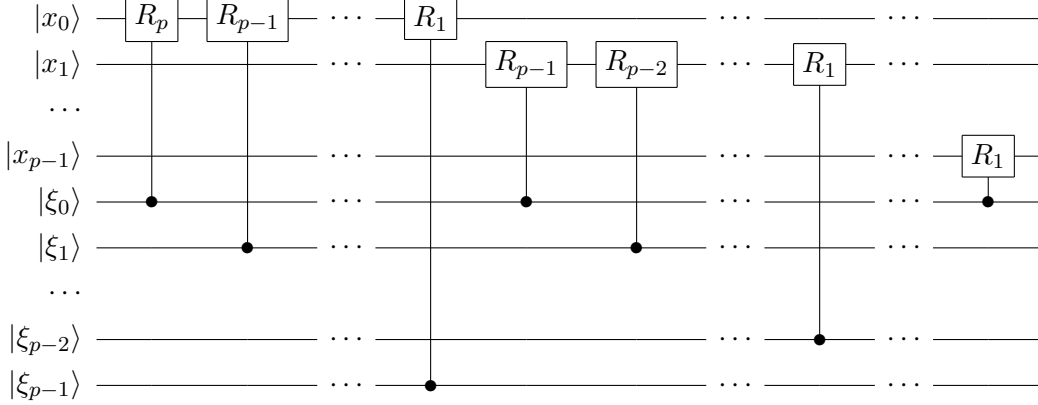
Then the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ is obtained by applying the Hadamard gates $H^{\otimes pd}$ to the $\mathbf{x}$ -register and putting both registers together.

Step 2. Multiply the phase $e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P}$ with U_{ph} . We develop an efficient implementation for multiplication without involving any extra error or ancilla qubits in the following lemma.

Lemma 2.1. *The following $2p$ -qubit circuit U_{ph} implements the unitary operator:*

$$|x\rangle |\xi\rangle \mapsto e^{2\pi i x \xi / P} |x\rangle |\xi\rangle, \quad 0 \leq x, \xi < P,$$

with $\mathcal{O}(p^2)$ gate complexity precisely without ancilla qubits.



Proof. We first write the binary representation of integers $x = (x_{p-1} \cdots x_0)$, $\xi = (\xi_{p-1} \cdots \xi_0)$, and do the following calculation:

$$\begin{aligned} e^{2\pi i x \xi / P} |x\rangle |\xi\rangle &= \prod_{j=0}^{p-1} \prod_{k=0}^{p-1} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{0 \leq j+k < p} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{k=0}^{p-1} e^{2\pi i x_0 \xi_k \cdot 2^{k-p}} |x_0\rangle \otimes \cdots \otimes \prod_{k=0}^0 e^{2\pi i x_{p-1} \xi_k \cdot 2^{k-1}} |x_{p-1}\rangle \otimes |\xi_{p-1} \cdots \xi_0\rangle. \end{aligned}$$

The circuit corresponding to the unitary can be implemented by a series of controlled rotations

$$\begin{array}{ccc} |x_j\rangle & \xrightarrow{R_{p-j-k}} & e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_j\rangle \\ | \xi_k \rangle & \xrightarrow{\bullet} & | \xi_k \rangle \end{array}$$

where $R_j = R_z(\pi/2^{j-1})$. Finally, the above circuit is obtained after arranging the controlled rotations in the corresponding places. $\square$

Rewriting the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ as

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i x_d \xi_d / P} \cdots e^{2\pi i x_1 \xi_1 / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle,$$

then the map from $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ to $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ can be performed by applying Lemma 2.1 for d times to the register pairs $(x_d, \xi_d), \dots, (x_1, \xi_1)$. The corresponding circuit is denoted as $U_{ph}^{\otimes d}$ and involves $\mathcal{O}(p^2 d)$ elementary gates with no ancilla qubits. After the multiplication of $U_{ph}^{\otimes d}$, one obtains the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$.

Step 3. Multiply the symbol $\check{a}(\mathbf{x}, \mathbf{x}i)$. The next component is the diagonal multiplication $U_{\check{a}}$, which is designed to approximate the map

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} \mapsto \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} + |\perp\rangle, \quad (5)$$

where $C_a > 0$ is a constant that depends on $\check{a}(\mathbf{x}, \boldsymbol{\xi})$ and b is the number of ancilla qubits used for $U_{\check{a}}$. As mentioned earlier, the idea is to treat $(\mathbf{x}, \boldsymbol{\xi})$ as a $2d$ -dimensional variable and utilize the result from arithmetic circuit construction. Leveraging the reversible computational model and the uncomputation technique, any classical arithmetic operation can be implemented by a quantum circuit efficiently. More specifically, one can construct a corresponding quantum circuit using $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}))$ ancilla qubits and $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}))$ gates, where ε is the precision one wants to achieve. We state a general result for an efficient block encoding of diagonal matrices D_g , as defined in (1), which is summarized in [Proposition 2.2](#).

Proposition 2.2. *Assume that $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is a smooth arithmetic function with $\sup|g| < +\infty$. Then there is an $(C, \mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(mp)), \varepsilon)$ -block-encoding of D_g with $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(mp))$ gates, where D_g is a diagonal operator on $\mathbb{C}^{mp}$ defined in (1), and $C \geq \sup|g|$.*

Proof. The circuit U_g is constructed as follows. Let $t = \lceil \log_2(\frac{C\pi}{\varepsilon}) \rceil$, and let $\theta(x_1, \dots, x_m)$ be a map that gives $|\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle$, where $(\theta_{t-1}\theta_{t-2}\dots\theta_0)$ is the closest t -bit fixed-point representation of $\frac{1}{\pi} \arcsin(|g(x_1, \dots, x_m)|/C)$, and $\theta_{\text{sgn}} = \mathbb{1}_{g < 0}$. For an arbitrary basis state $|x_m\rangle \dots |x_1\rangle$, we consider the system with $t + 2$ ancilla qubits $|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1}$. Here $|x_j\rangle = |x_{j,p-1} \dots x_{j,0}\rangle, \forall j$. Using the reversible computational model and uncomputation, the classical circuit:

$$|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1} \mapsto |0\rangle |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle$$

can be constructed with $\mathcal{O}(\text{poly}(t) + \text{poly}(mp))$ gates and $\mathcal{O}(\text{poly}(mp))$ ancilla qubits. Then we apply the following circuit on the $t + 2$ ancilla qubits. The state obtained is:

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1}\dots\theta_0)) |1\rangle + \sin(\pi(\theta_{t-1}\dots\theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}}\theta_{t-1}\theta_{t-2}\dots\theta_0\rangle,$$

which can then be mapped to

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1}\dots\theta_0)) |1\rangle + \sin(\pi(\theta_{t-1}\dots\theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1},$$

via uncomputation.

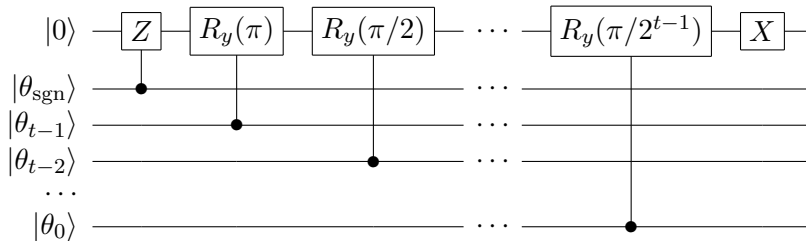
Since

$$\left| (\theta_{t-1}\dots\theta_0) - \frac{1}{\pi} \arcsin(|g(x_1, \dots, x_m)|/C) \right| < \frac{\varepsilon}{C\pi},$$

we have

$$\left| (-1)^{\theta_{\text{sgn}}} \sin(\pi(\theta_{t-1}\dots\theta_0)) - \frac{1}{C} g(x_1, \dots, x_m) \right| < \frac{\varepsilon}{C}$$

which means the desired state $\frac{1}{C} g(x_1, \dots, x_m) |x_m\rangle \dots |x_1\rangle$ is obtained with error at most $\frac{\varepsilon}{C}$ upon measuring the ancilla qubits and getting $|0\rangle^{\otimes t+2}$.



□

As a result of applying the block encoding given in [Proposition 2.2](#) with $m = 2d$, $g = \check{a}$ and ε replaced by $\varepsilon/\sqrt{P^d}$ to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b}$, one obtains the state $|\phi\rangle |0\rangle^{\otimes b} + |\perp\rangle$ using $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(pd))$ ancilla qubits and $\mathcal{O}(\text{polylog}(\frac{1}{\varepsilon}) + \text{poly}(pd))$ gates where

$$\left\| |\phi\rangle - \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle \right\| < \frac{\varepsilon}{\sqrt{P^d} C_a}, \quad (6)$$

and $C_a \geq \sup |a| = \sup |\check{a}|$ is a constant. Here $|\phi\rangle$ denotes an unnormalized $2pd$ -qubit state.

Step 4. Sum over the frequency variable. Finally, after applying the Hadamard gate $H^{\otimes pd}$ to the $\boldsymbol{\xi}$ registers, we obtain the state

$$(I_{pd} \otimes |0\rangle^{\otimes pd})(I_{pd} \otimes \langle 0|^{\otimes pd})(I_{pd} \otimes H^{\otimes pd}) |\phi\rangle |0\rangle^{\otimes b} + |\perp'\rangle,$$

where

$$\left\| (I_{pd} \otimes |0\rangle^{\otimes pd})(I_{pd} \otimes \langle 0|^{\otimes pd})(I_{pd} \otimes H^{\otimes pd}) |\phi\rangle |0\rangle^{\otimes b} \right. \quad (7)$$

$$\left. - \frac{1}{\sqrt{P^d} \|f\| C_a} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |0\rangle^{\otimes pd} |0\rangle^{\otimes b} \right\| < \frac{\varepsilon}{\sqrt{P^d} C_a}, \quad (8)$$

which can be seen from (6). Therefore, one obtains the desired state on the $\mathbf{x}$ registers upon measuring $|0\rangle^{\otimes pd}$ for the $\boldsymbol{\xi}$ registers and $|0\rangle^{\otimes b}$ for the ancilla qubits. Notice that there is an extra scaling factor $\frac{1}{\sqrt{P^d}}$ due to the application of Hadamard gates, so the success probability is $\mathcal{O}(2^{-pd})$.

The block encoding scheme of the PDO (3) constructed in this section can be summarized in the following theorem:

Theorem 2.3. *For a generic symbol $a(\mathbf{x}, \boldsymbol{\xi})$, a block encoding of the corresponding discretized PDO defined by (3) can be $(2^{\frac{pd}{2}} C_a, \mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon})), \varepsilon)$ -block-encoded using the above circuit with gate complexity $\mathcal{O}(\text{poly}(pd) + \text{polylog}(\frac{1}{\varepsilon}))$, where $C_a \geq \sup |a|$ is a constant.*

3 Efficient block encoding for separable symbols